The Phase 1 Constraint-Projection QP and its Concave Dual

Exact Euclidean projection onto partition, equal-area, and box constraints: dual derivation, the per-vertex simplex solve, the outer Jacobian, and the exactness/idempotency guarantees

July 2026

Status — specification (pre-implementation)

This document is the mathematical *source of truth* for the planned function `orthogonal_projection_newton` in `src/optimization/projection.py` (design: `docs/plans/PHASE1_DUAL_NEWTON_PROJECTION_PLAN.md`). Unlike the other `docs/math/` documents, it is written *before* the code (the plan's workflow: derive first, then implement). The code call-out boxes therefore name a function that does not yet exist; they define what it must compute. The measured motivation — that the current `orthogonal_projection_iterative` does *not* compute this projection — is documented in `docs/experiments/03-dual-projection-verification/`.

Abstract

Each Phase 1 PGD iteration projects a candidate density onto the intersection of the partition constraint ($\sum_k u_{i,k} = 1$ per vertex), the equal-area constraint ($\sum_i v_i u_{i,k} = d_k$ per cell), and the box $0 \le u \le 1$. This document derives the exact Euclidean projection through its Lagrangian dual. The primal is recovered from the duals by a clip ((3)); eliminating the per-vertex row duals is a probability-simplex projection, which must be computed *cap-free* for the row dual to be well defined (§4). What remains is the maximization of a concave, C^1, piecewise-quadratic *partial dual* $q(\beta)$ of dimension N (the cell count), with $\nabla q = -R$ the negated area residual (§5). We derive its generalized Hessian $-J$ (the $N \times N$ area Jacobian, (11)), prove it is symmetric positive semidefinite with the *structural* kernel $\operatorname{span}\{\mathbf{1}\}$ (§7), and use this to justify the solver: globally-convergent L-BFGS on q (primary), with a gauge-fixed semismooth-Newton polish (§8). Finally we prove exactness and idempotency (§9). All quantities are analytical.

Contents

1 Overview

Phase 1 minimizes a Γ-convergence energy by projected gradient descent on a nodal density $U \in \mathbb{R}^{V\times N}$ (V mesh vertices, N cells). Each step takes a post-gradient candidate Y (generally infeasible) and returns the nearest feasible density — the Euclidean projection onto the constraint set. This projection dominates Phase-1 wall time and, as measured in `docs/experiments/03-dual-projection-verification/`, the incumbent iterative routine does not actually compute it. Here we derive the exact projection and the structure that makes it cheap: a per-vertex closed form plus a small N-dimensional concave maximization.

2 Notation and Setup

$U = (u_{i,k}) \in \mathbb{R}^{V\times N}$ is the density: row i is vertex i, column k is cell k. $v \in \mathbb{R}^V$ is the lumped P_1 mass (positive), $\mathbf{1}$ a vector of ones (length inferred), $\Delta = \{p \in \mathbb{R}^N : \sum_k p_k = 1,\ p \geq 0\}$ the probability simplex. Given the candidate Y and per-cell area targets $d \in \mathbb{R}^N$ ($d_k = \frac{1}{N}\sum_i v_i$ in the equal-area case), the projection is

$$\min_U \ \tfrac{1}{2}\|U - Y\|^2 \quad \text{s.t.} \quad \underbrace{\textstyle\sum_k u_{i,k} = 1 \ \ \forall i}_{\text{row / partition}}, \quad \underbrace{\textstyle\sum_i v_i u_{i,k} = d_k \ \ \forall k}_{\text{area}}, \quad 0 \leq u_{i,k} \leq 1. \tag{1}$$

The metric is the plain (unweighted) Euclidean norm — verified against the incumbent's correction term in `docs/experiments/03-...` and §3.

Remark 2.1 (the upper box bound is redundant)**.** If $\sum_k u_{i,k} = 1$ and $u_{i,k} \geq 0$ then each $u_{i,k} \leq 1$ automatically. Hence the box in (1) reduces to $U \geq 0$, and the per-vertex feasible set is exactly Δ. The cap at 1 never binds; it is nonetheless important *not* to carry it inside the inner solve (§4, Remark 4.1).

Remark 2.2 (feasibility of (1))**.** Summing the row constraints weighted by v gives $\sum_k\sum_i v_i u_{i,k} = \sum_i v_i$, so the area targets must satisfy the consistency condition $\sum_k d_k = \sum_i v_i$. Under it (and $d \geq 0$) the constant-column density $u_{i,k} = d_k/\sum_i v_i \in (0,1)$ is strictly feasible, so (1) is feasible and Slater's condition holds — strong duality applies.

3 The dual and the primal-from-dual map

Introduce a multiplier α_i for each row constraint and β_k for each area constraint. Keeping the box $0 \leq U \leq 1$ explicit, the Lagrangian is

$$\mathcal{L}(U,\alpha,\beta) = \tfrac{1}{2}\|U - Y\|^2 - \sum_i \alpha_i\Big(\sum_k u_{i,k} - 1\Big) - \sum_k \beta_k\Big(\sum_i v_i u_{i,k} - d_k\Big). \tag{2}$$

The dual function is $g(\alpha,\beta) = \min_{0\leq U\leq 1} \mathcal{L}(U,\alpha,\beta)$. Because (2) separates over entries, the minimizing U is obtained entrywise: the unconstrained stationary point is $y_{i,k} + \alpha_i + v_i\beta_k$, clipped to the box, giving the **primal-from-dual map**

$$u_{i,k}(\alpha,\beta) = \mathrm{clip}\big(y_{i,k} + \alpha_i + v_i\beta_k,\ 0,\ 1\big). \tag{3}$$

Note the row dual α_i enters with unit weight and the area dual β_k with the mass weight v_i — the signature of the *Euclidean* (not v-weighted) metric. Substituting (3) into (2),

$$g(\alpha,\beta) = \sum_{i,k}\Big[\tfrac{1}{2}\big(u_{i,k} - y_{i,k}\big)^2 - (\alpha_i + v_i\beta_k)\,u_{i,k}\Big] + \sum_i \alpha_i + \sum_k \beta_k d_k, \tag{4}$$

a concave function of (α, β) (a pointwise minimum, over U, of maps that are affine in (α, β)). By Danskin's theorem [1], with $u = u(\alpha, \beta)$ optimal,

$$\frac{\partial g}{\partial \alpha_i} = 1 - \sum_k u_{i,k}, \qquad \frac{\partial g}{\partial \beta_k} = d_k - \sum_i v_i u_{i,k} =: -R_k(\alpha, \beta), \tag{5}$$

where $R \in \mathbb{R}^N$ is the *area residual.* The projection is $U(\alpha^\star, \beta^\star)$ at the dual maximizer, where both gradients vanish (feasibility) and (3) holds (KKT); by Remark 2.2 strong duality makes this exact.

Code: primal reconstruction

`src/optimization/projection.py`: `orthogonal_projection_newton` (planned).
The final primal is `U = clip(Y + alpha[:,None] + v[:,None]*beta[None,:], 0, 1)` ((3)); `alpha`, `beta` come from §4–§8.

4 Eliminating the row duals: the per-vertex simplex solve

For fixed β, (4) separates over vertices, so each α_i is found independently by maximizing g in α_i, i.e. by solving $\partial g / \partial \alpha_i = 0$ from (5). Writing $z_{i,k} := y_{i,k} + v_i \beta_k$,

$$\sum_k \operatorname{clip}\big(z_{i,k} + \alpha_i,\ 0,\ 1\big) = 1. \tag{6}$$

By Remark 2.1 this is the projection of z_i onto the probability simplex Δ: the minimizer of $\frac{1}{2}\|u_i - z_i\|^2$ over $u_i \in \Delta$ has the form $u_{i,k} = \max(z_{i,k} - \tau_i, 0)$ with the threshold $\tau_i = -\alpha_i$ fixed by $\sum_k u_{i,k} = 1$. This is Condat's exact finite (sort/threshold) solve [3].
Proposition 4.1 (well-posedness of the row dual)**.** *Solve* (6) *cap-free, i.e. as projection onto* $\Delta = \{\sum = 1,\ \cdot \geq 0\}$*. Then* α_i *is the unique root of* $h_i(\alpha) := \sum_k \max(z_{i,k} + \alpha, 0) = 1$*.*

Proof. h_i is continuous, nondecreasing, convex and piecewise linear, with $h_i \to 0$ as $\alpha \to -\infty$ and $h_i \to +\infty$ as $\alpha \to +\infty$. On any interval where at least one argument is positive its slope equals the number of active entries ≥ 1, so h_i is *strictly* increasing there. Since the level $1 > 0$ is attained only where some entry is active, the root is unique. □

Remark 4.1 (cap-free is a well-posedness requirement, not an optimization)**.** If instead the cap is retained inside (6), $\tilde{h}_i(\alpha) = \sum_k \operatorname{clip}(z_{i,k} + \alpha, 0, 1)$ has *flat* segments wherever entries saturate at 1: whenever a row's largest argument exceeds the runner-up by more than 1, the level 1 falls on such a plateau and (6) has an interval of solutions in α_i. E.g. $z_i = (2.5, -0.3, -0.8)$ gives $\tilde{h}_i(\alpha) = 1$ for every $\alpha \in [-1.5, 0.3]$. Then $\alpha_i(\beta)$ is set-valued and $d\alpha_i/d\beta$ — needed for the outer Jacobian (§7) — is undefined. The cap-free solve returns the unique $\alpha_i = 1 - \max_k z_{i,k}$ (here -1.5) and the identical primal $u_i = (1, 0, 0)$ (Remark 2.1). The V vertices are independent, so the whole pass is a single vectorized sort/threshold; it is *stateless*, recomputed from β each outer step.

Code: inner solve

`src/optimization/projection.py`: `orthogonal_projection_newton` (planned).
Vectorized cap-free simplex projection of `Z = Y + v[:,None]*beta[None,:]` onto Δ (sort each row, find τ_i, `U = maximum(Z - tau[:,None], 0)`); `alpha = -tau`. Never carry `clip(.,0,1)` into this solve (Remark 4.1).

5 The concave partial dual $q(\beta)$

Let $\alpha(\beta)$ denote the row duals of §4 and define the **partial dual**

$$q(\beta) := \max_{\alpha} \; g(\alpha, \beta) = g\big(\alpha(\beta), \beta\big). \tag{7}$$

Proposition 5.1 (*q is concave, C^1, with $\nabla q = -R$*)**.** *q is concave and continuously differentiable, and*

$$\nabla q(\beta) = -R(\beta), \qquad R_k(\beta) = \sum_i v_i \, u_{i,k}\big(\alpha(\beta), \beta\big) - d_k, \tag{8}$$

where $u(\alpha(\beta), \beta)$ is the primal (3). Maximizing q over β solves (1).

Proof. g is jointly concave (§3); partial maximization over the convex variable α preserves concavity, so q is concave. The inner maximizer $\alpha(\beta)$ is unique (Prop. 4.1) and Lipschitz in β; by Danskin's theorem [1] q is differentiable with $\nabla q(\beta) = \partial g/\partial\beta\big|_{\alpha=\alpha(\beta)} = d - \sum_i v_i u_{i,\cdot} = -R$ (using $\partial g/\partial \alpha = 0$ at $\alpha(\beta)$). The residual R is continuous (the simplex projection is 1-Lipschitz), so $q \in C^1$. At $\beta^\star = \arg\max q$, $R(\beta^\star) = 0$ and the inner solve enforces the row constraints, so $u(\alpha(\beta^\star), \beta^\star)$ is primal feasible and satisfies (3) — the KKT conditions of (1); strong duality (Remark 2.2) makes it the projection. $\square$

Consequence for the solver. The outer problem is the maximization of a concave C^1 function of only N variables, whose value and gradient cost one vectorized inner pass. This is what makes the projection cheap and robust (§8).

6 The gauge and the structural singularity

Proposition 6.1 (shift invariance / gauge)**.** *For any $t \in \mathbb{R}$, $(\alpha, \beta) \mapsto (\alpha - t\, v, \;\; \beta + t\,\mathbf{1})$ leaves every $u_{i,k}$ in (3) unchanged. Consequently $q(\beta + t\mathbf{1}) = q(\beta)$, $\nabla q \cdot \mathbf{1} \equiv 0$, and $\mathbf{1}^\top R \equiv 0$ after the inner solve.*

Proof. The argument of the clip is $y_{i,k} + \alpha_i + v_i \beta_k$; under the shift it becomes $y_{i,k} + (\alpha_i - t v_i) + v_i(\beta_k + t) = y_{i,k} + \alpha_i + v_i \beta_k$, unchanged, so u (hence g) is invariant. As $\beta \mapsto \beta + t\mathbf{1}$ is matched by $\alpha(\beta + t\mathbf{1}) = \alpha(\beta) - tv$, $q(\beta + t\mathbf{1}) = g(\alpha(\beta), \beta) = q(\beta)$; differentiating in t gives $\nabla q \cdot \mathbf{1} = 0$, i.e. $\mathbf{1}^\top R = 0$. Directly: $\mathbf{1}^\top R = \sum_i v_i \big(\sum_k u_{i,k}\big) - \sum_k d_k = \sum_i v_i - \sum_k d_k = 0$ by Remark 2.2 and $\sum_k u_{i,k} = 1$. $\square$

Thus the dual maximizer is a line $\beta^\star + \operatorname{span}\{\mathbf{1}\}$; the *primal* u is unique on it. Any Newton-type linear solve must fix this gauge (e.g. impose $\sum_k \beta_k = 0$, or drop one component); L-BFGS simply drifts harmlessly along the ridge. The residual identity $\mathbf{1}^\top R = 0$ means the Newton system of §7 is always consistent on $\mathbf{1}^\perp$.

7 The outer Jacobian $J = -\nabla^2 q$

Away from the finitely many β where an entry crosses a box breakpoint, the active sets

$$A_i = \{k : \;\; z_{i,k} + \alpha_i > 0\}, \qquad m_i := |A_i| \geq 1, \tag{9}$$

are locally constant, and (cap-free, Prop. 4.1) $u_{i,k} = (z_{i,k} + \alpha_i)\,[k \in A_i]$ with $\alpha_i = (1 - \sum_{k \in A_i} z_{i,k})/m_i$. Since $\partial z_{i,k}/\partial \beta_l = v_i \delta_{kl}$,

$$\frac{\partial \alpha_i}{\partial \beta_l} = -\frac{v_i}{m_i}\,[l \in A_i], \qquad \frac{\partial u_{i,k}}{\partial \beta_l} = \Big(v_i \delta_{kl} - \frac{v_i}{m_i}[l \in A_i]\Big)[k \in A_i]. \tag{10}$$

Differentiating $R_k = \sum_i v_i u_{i,k} - d_k$ gives the generalized Jacobian

$$\boxed{J_{kl} = \frac{\partial R_k}{\partial \beta_l} = \delta_{kl} \sum_{i:\, k\in A_i} v_i^2 - \sum_{i:\, k,l\in A_i} \frac{v_i^2}{m_i}}, \qquad J = \sum_i v_i^2 \Big(\mathrm{diag}(a_i) - \tfrac{1}{m_i} a_i a_i^\top\Big), \tag{11}$$

with $a_i \in \{0,1\}^N$ the indicator of A_i ($m_i = a_i^\top a_i$). Since $R = -\nabla q$, $J = -\nabla^2 q$.

Proposition 7.1 (structure of J)**.** *J is symmetric positive semidefinite, and* $J\mathbf{1} = 0$ structurally *(for every active-set configuration). Its kernel is $\{x : x$ constant on each $A_i\}$; if the active sets jointly connect all N cells this is exactly* $\mathrm{span}\{\mathbf{1}\}$ *and J has rank $N-1$.*

Proof. Each summand $B_i = \mathrm{diag}(a_i) - \frac{1}{m_i} a_i a_i^\top$ is symmetric, and for any x, $x^\top B_i x = \sum_{k\in A_i} x_k^2 - \frac{1}{m_i}\big(\sum_{k\in A_i} x_k\big)^2 \geq 0$ by Cauchy–Schwarz, with equality iff x is constant on A_i. Hence $J = \sum_i v_i^2 B_i \succeq 0$. Also $B_i a_i^{(\mathbf{1})}$ where $B_i\mathbf{1} = a_i - a_i(a_i^\top \mathbf{1})/m_i = a_i - a_i = 0$, so $J\mathbf{1} = 0$ regardless of the active sets. The kernel is $\bigcap_i \{x \text{ const on } A_i\}$; connectedness collapses it to the constants. □

The only linear system anywhere in the method is the *gauge-fixed* reduced $(N-1)\times(N-1)$ solve of §8; N is the cell count ($\lesssim 10^3$), so it is a trivial dense factorization.

> **Code: outer gradient / Jacobian**
>
> `src/optimization/projection.py`: `orthogonal_projection_newton` (planned).
> Gradient `-R` from one inner pass ((8)); Jacobian (11) only if the Newton polish is used, assembled from the active-set indicators a_i and solved on $\{\sum \beta = 0\}$ (never an ε-ridge — see §8).

8 Solving the outer problem

Primary — L-BFGS on the concave dual. Maximize q (equivalently minimize $-q$) by L-BFGS-B [2]; each evaluation is one inner pass (§4) returning q and $\nabla q = -R$. Because q is concave and C^1 (Prop. 5.1), L-BFGS with a Wolfe line search converges globally to the maximizer (unique modulo the $\mathbf{1}$-gauge, Prop. 6.1); the primal is unique. No Jacobian, no linear solve, no active-set bookkeeping.

Optional polish — semismooth Newton on the gauge-fixed quotient. R is semismooth (piecewise smooth in β); the Newton step solves $J\,\delta\beta = -R$ for $\delta\beta \in \mathbf{1}^\perp$ (consistent by $\mathbf{1}^\top R = 0$). Since $J \succeq 0$, $\delta\beta = -J^+ R$ is an ascent direction for q ($\nabla q \cdot \delta\beta = R^\top J^+ R \geq 0$), so use q itself as the line-search merit. On the quotient $\{\sum \beta = 0\}$, if the active sets connect all cells, J is positive definite (Prop. 7.1) and the generalized Jacobians at $\beta^\star$ are nonsingular (BD-regularity), so Qi–Sun [5, 4] gives local quadratic convergence there. A handful of Newton steps drive feasibility from the L-BFGS tolerance to machine level.

Remark 8.1 (why not an ε-ridge; the empty-cell degeneracy)**.** The structural kernel $\mathrm{span}\{\mathbf{1}\}$ must be removed by the reduced/gauge-fixed solve, *not* by a Tikhonov ridge $J + \varepsilon I$. The ridge is not merely inelegant: if some cell k has *no* active entry at any vertex (an empty cell), then row k of J is zero while $R_k = -d_k \neq 0$, so $J\,\delta\beta = -R$ is inconsistent in coordinate k; the ridge returns $\delta\beta_k = d_k/\varepsilon$ (an explosion, $\sim 10^{10}$ at $\varepsilon = 10^{-4}$), while a least-squares solve leaves $\delta\beta_k = 0$ and never revives the cell. The concave ascent has no such failure: $\partial q/\partial\beta_k = -R_k = d_k > 0$ drives β_k up until cell k re-enters some active set. Hence L-BFGS is the robust primary; the Newton polish must detect empty active columns and fall back to an ascent step there.

Code: outer solve

`src/optimization/projection.py`: `orthogonal_projection_newton` (planned).
Primary: `scipy.optimize.minimize(-q, beta0, jac=..., method="L-BFGS-B")`, warm-started on β from the previous PGD step. Optional Newton polish: gauge-fixed reduced solve of (11) with q-ascent line search and the empty-cell safeguard (Remark 8.1). Thread only β (α is recomputed).

9 Exactness and idempotency

Proposition 9.1 (exactness and idempotency)**.** *At the dual maximizer $\beta^\star$, $U^\star = U(\alpha(\beta^\star), \beta^\star)$ from (3) is the exact solution of (1). Moreover the projection is idempotent: if Y already satisfies (1), then $U^\star = Y$ to machine precision.*

Proof. Exactness. At $\beta^\star$, $\nabla q(\beta^\star) = -R(\beta^\star) = 0$ (area feasibility) and the inner solve enforces $\sum_k u^\star_{i,k} = 1$ (row feasibility) with $u^\star \geq 0$; (3) is the box-KKT stationarity with multipliers $(\alpha(\beta^\star), \beta^\star)$. These are the KKT conditions of the convex program (1), which are sufficient; strong duality (Remark 2.2) identifies $U^\star$ with the projection. *Idempotency.* If Y is feasible then $\beta = 0$, $\alpha = 0$ give $R = 0$ and reproduce Y exactly, so $\beta^\star = 0$ is a maximizer and $U^\star = \mathrm{clip}(Y, 0, 1) = Y$. A warm-started solve terminates at once, with movement at the level of the solver tolerance. ☐

This is precisely what the incumbent lacks: it is not the projection at all (gap up to 0.86; `docs/experiments/03-dual-projection-verification/`), and its clip∘project step is only weakly idempotent ($\sim 6 \times 10^{-8}$ per iteration). The dual projection removes both.

Summary table

Quantity	Form	Where
Primal from duals	$u_{i,k} = \mathrm{clip}(y_{i,k} + \alpha_i + v_i\beta_k, 0, 1)$	(3), analytical
Row dual $\alpha_i(\beta)$	cap-free simplex projection of z_i (unique)	§4, analytical
Partial dual $q(\beta)$	concave, C^1, $\nabla q = -R$	Prop. 5.1, analytical
Gauge	$(\alpha, \beta) \to (\alpha - tv, \beta + t\mathbf{1})$; $\nabla q \cdot \mathbf{1} = 0$	Prop. 6.1
Outer Jacobian $J = -\nabla^2 q$	$\delta_{kl} \sum_{i:k\in A_i} v_i^2 - \sum_{i:k,l\in A_i} v_i^2/m_i$	(11), analytical
J structure	sym. PSD, $J\mathbf{1} = 0$, rank $N-1$	Prop. 7.1
Primary solver	L-BFGS on q (global)	§8
Polish	semismooth Newton on $\{\sum \beta = 0\}$ (quadratic)	§8
Guarantee	exact projection; idempotent	Prop. 9.1

References

[1] Dimitri P. Bertsekas. *Nonlinear Programming.* Athena Scientific, Belmont, MA, 2nd edition, 1999. Lagrangian duality for box/equality-constrained QP, Danskin's theorem (gradient of a partial-min value function), and concavity of the dual.

[2] Richard H. Byrd, Peihuang Lu, Jorge Nocedal, and Ciyou Zhu. A limited memory algorithm for bound constrained optimization. *SIAM Journal on Scientific Computing*, 16(5):1190–1208, 1995. L-BFGS-B; the primary outer solver maximizing the concave partial dual (available as scipy.optimize.minimize method=L-BFGS-B).

[3] Laurent Condat. Fast projection onto the simplex and the ℓ_1 ball. *Mathematical Programming*, 158(1–2):575–585, 2016. Exact finite (sort/threshold) projection onto the probability simplex; the per-vertex inner solve of the dual projection.

[4] Michael Hintermüller, Kazufumi Ito, and Karl Kunisch. The primal-dual active set strategy as a semismooth newton method. *SIAM Journal on Optimization*, 13(3):865–888, 2002. Box-constrained quadratic programs solved as a semismooth Newton method via the active-set structure used in the Newton polish.

[5] Liqun Qi and Jie Sun. A nonsmooth version of newton's method. *Mathematical Programming*, 58(1–3):353–367, 1993. Semismooth Newton method and its local quadratic convergence under BD-regularity; basis for the optional Newton polish of the dual projection.